🧠 D503 ENGINE - EVERY WEIGHT

How the D503 Sports Engine picks - every weight, every rule, in 12 parts.

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1/12 🧠 THE CORE

- Power rating for every team (Elo), updated after every game - margin of victory counts, and ratings slide 1/3 back to average over the offseason. 122,000+ finals across six sports, 10+ years.
- Home edge (rating pts, re-picked every day): NFL 40, college FB 60, NBA 40, NHL 20, MLB 40. College hoops' rating settings are being re-tuned this week.
- On top, the model weighs 14 factors: recent form, rest gap, back-to-backs, injuries, QB / goalie / starting pitcher form, revenge, letdown spots, byes, short weeks, overseas games, altitude, cold, nasty weather, time zones crossed. Weights refit every day on the last 8,000 games.
- That's the engine's OWN line. Then it blends with the books' price - trusting its own read only as much as it has beaten Vegas (learned per sport): MLB 34%, NFL 23%, NBA 2%, college FB 0%. Hockey = the line + only back-to-back and rest (the two things the hockey study proved the line misses).
- Two more corrections: its real track record per sport (the top of its reads trimmed - NBA and college FB about 3.7 pts, NFL and MLB about 1; NHL and college hoops nudged up under 1 - never below the line), and the big price study where the books' own price is proven off (NHL, college hoops).
- Hoops opening weeks: the read leans on the books until both teams have 10 games (the NBA starts halfway, thanks to the roster read).
- Then the study weights go on top: all of them together max ±6 pts on a dog, ±4 on a favorite. A pick only happens when the WHOLE picture beats the price - never one factor.

2/12 📚 WHAT THE MODEL LEARNED EACH FACTOR IS WORTH

- How to read it: + = the factor really helps the side it points to. - = it's over-hyped (that side does worse than you'd think). Refit daily - these are the 11/10 numbers. Tiny ones left out.
- 🏈 NFL (home 40, trust 23%): rating .95 | cold +.31 | QB form +.27 | altitude +.26 | time zones -.26 | bye +.17 | hot streak -.17 | injuries +.15 | letdown -.11 | revenge +.11 | rest +.09 | short week +.09 | weather +.06
- 🏈 COLLEGE FB (home 60, trust 0% - the raw ratings read only drives the 🔍 play): rating 1.22 | overseas -.33 | QB form +.31 | time zones -.28 | injury count -.20 | rest -.18 | bye -.17 | letdown -.15 | altitude +.14 | short week -.12
- 🏀 NBA (home 40, trust 2% - NBA favorites read off the books, part 7): rating .87 | hot streak -.39 | back-to-back +.26 | rest +.22 | injuries +.17 | altitude +.15 | letdown -.10 | time zones -.09
- 🏒 NHL (home 20): the live read = the line + back-to-back .13 + rest .01. The full model (used only when there's no line): rating 1.02 | injuries +.36 | back-to-back +.31 | hot streak -.24 | rest +.19 | overseas +.19 | letdown +.09
- ⚾ MLB (home 40, trust 34%): rating .99 | injuries +.27 | back-to-back +.10 | starting pitcher +.09 | rest -.09 | cold -.06 | revenge +.06 | weather +.05 | time zones -.05
- 🏀 COLLEGE HOOPS: its rating weights are being re-tuned this week. Its unit plays run on separate efficiency ratings (part 8).
- What it says: hot streaks are over-priced (NFL, NBA, NHL), blowout winners keep rolling (NFL, NBA), and travelers do better than the hype.

3/12 🚑 INJURIES & KEY PLAYERS (EVERY SPORT)

- Full injury data, every team: ESPN for the pros, the official availability reports for college (pulled 5:47 and 11:47 AM PT). No data on a team = UNKNOWN, never 'healthy' = no pick.
- The starting QB (started the last game), a starting goalie, an NBA starter (3+ of the last 10) or an MLB star bat OUT: the ratings can't see it, so that game

goes by the books' price (it already priced the backup); the spread by the book's own line. Baseball waits for both starting pitchers.

- Key men out take points off that side's own read (box-score studies 2021-26): NFL QB -8, lead RB -3, top-2 WR -3, two+ -10 | college QB -3, two+ -5 | NHL top scorer -6, a top-2 -4, two of the top 3 -7 (cap 8) | NBA top scorer -9, two of the top 3 -11 (cap 12) | MLB none (noise). One man counts once.
- 11/4 fix: a college key man who already missed the last game is already in the price (+0.1 on 1,398 vs -3.0 for a fresh one) - no points. The NFL keeps it (-3.1 on 400, 5 of 6 seasons).
- Football 'regulars' = the QB, top 2 ball carriers, top 4 catchers, top 11 tacklers in the last 3 box scores. The more banged-up side: NFL -½ pt per extra regular out, cap 3; college 0 (2+ out doesn't lose vs the price). Weighed, never a block.
- NBA: only a KEY man (top-3 scorer or a regular starter) holds a pick. A guy who plays through tags (10+ times questionable, 85%+ played) counts as playing, with 1-in-5 of his points weighed. Any other key questionable = the pick waits (54.5% of 6,380 played).
- Hockey, college hoops, MLB (and any team with no box scores): 2+ out or 4+ questionable = no units, a lean at most (not studied yet). Hockey only counts guys who played 30%+ of the last 10.
- A unit play never posts while it waits on news. Key men missing on the other side are a named reason; the card names who's out 🚑.

4/12 🐶 THE DOG SCORE & THE FAVORITE READ

- Dog score = the engine's own read minus the books' no-vig price (in points), then every study angle added. The own read counts at most +12; in the NFL and NBA a read past +12 counts ZERO (those big reads were traps - 43k-dog study).
- All the angles together: max ±6 (overlapping spots never outweigh the read).
- Angles in every sport: money ran 2+ pts away since the open -4 (dogs lost: NFL -40%, college -9%, NBA -8%, NHL -6%) | rested vs a team on a back-to-back +3 (NBA, NHL) | its goalie / stars much hotter -3 (books over-rate it) |

off a blowout loss the market overreacts to +3 football, +5 NBA (upset as a -250 favorite) | a dog again after a +200 upset win -3 (NFL, college, MLB) | just fired its coach -3 football & hoops, +2 hockey | playoffs: the favorite lost the last series game +3.

- THE DOG GATE: NFL 8+, NHL 6+, college FB 4+, NBA 4+. No college hoops Dog of the Day (every replay lost).
- Own-read rule: football - its own read beats the real price; hockey - not under the no-vig number; NBA - none (the score decides). A 'trap' (a price spot the books are proven to overprice, like NHL road dogs on a back-to-back) never goes.
- Dog of the Day = the best score among real-value dogs +100..+350 (score over 0, read beats the price). A +200 or longer only if it's 5+ pts more value than the best shorter dog and reads 22%+.
- THE FAVORITE READ: every favorite moves the OTHER way - ½ pt per point of study angles on the dog across, cap ±4. Hockey does it off the no-vig line, the NBA off the books' price plus its own list. A favorite plays only if the whole weighed read beats its price.
- Spread sides get the same studies as small nudges on their rank - leans only (unit plays are moneyline only).

5/12 🏈 NFL

- Dog gate 8 (8+ went +10.0% on 212). The dog score alone isn't a named reason in football - no-reason dogs lost about -8% in the replays.
- Dog angles: 0-2 team +3 (blind 11-9, +76%) | East Coast team out West +3 (+20%, 5 of 6 seasons) | off a bye vs a team that played +3 (+26.7% on 49) | off a 17+ win +2 | the favorite lost its last +2 | neutral site the engine likes +1.5 | Monday night +1 (halved - +3.3% on 80 in the band) | scored 10 or fewer last game +1 | 3+ game win streak +1 | coach goes for it on 4th more / less ±1
- Dog fades: we lost and they won -3 | ice cold (last 3 games 7+ worse than its season) -3 | coach's first season with the team -3 | playing without its QB -3 (QB-out-again dogs -54% on 57)

- Favorite: a West Coast team favored on the road in an Eastern city +2 (won 76%, 7 of 8 seasons); the Eastern home dog -2.
- QB form = yards a throw with TDs and picks, his last 4 starts this season. QB out: the read goes to the price; key men -8 / -3 / -3 / -10; depth -½ per extra regular out (cap 3).
- Lock = 1u in the NFL (football Locks lost at bigger sizes).
- 🏈 Monday & Thursday: every NFL game gets its own pick (a lean is fine; a unit pick still waiting on news drops to a lean).
- Early: off a bye 1u, Monday night dog ½u, East Coast flying West ½u, hammered ½u, Europe morning under ½u (part 10). 'Blew somebody out' is OFF in the NFL (blind -12.5%).
- The scoreboard-beating-its-yards angle is tracked only in the NFL (no weight).

6/12 🏈 COLLEGE FOOTBALL

- Its read = the books' price, shaded (trust 0 - the college ratings hold nothing the market lacks). The raw ratings read drives the 🔍 play only.
- Never false info: a team whose games we don't ALL hold gets no record / streak line, no early spot, and the pick check holds it.
- Dog gate 4 (4+ went +18.9% on 228 when built).
- Dog angles: interception-unlucky (-0.5 a game or worse) +2 (+20.4% on 277) | off a 17+ win +2 | off a bye +2 | the favorite lost its last +2 | .700+ team +1.5 | neutral site the engine likes +1.5
- Dog fades: scoreboard 6+ a game over its yards -1 (-12.7% on 328, 6 of 6 seasons) | beat the spread by 21+ last game -2 | conservative-4th-down or fast-pace coach -2 | we lost and they won -3 | ice cold -3, 3+ game skid -3 (-4 for both) | without its QB -3 | under .300 after 4+ games -2
- Favorite (cap 4): off a loss -3 (-101..-150, 8 of 9 seasons) | RULE A +3 (-101..-150, off a win vs a team off a loss, the better-rated team: 9 of 9 seasons, blind 60-29) - a weight, never automatic | on its 3rd+ QB this season -2 (-101..-250, FBS v FBS; -27.4% on 73)

- 🔍 THE ENGINE SEES SOMETHING: a dog +100..+350 whose ratings read (injuries in) is 15+ pts over the books, the ratings themselves backing the dog (not a gap made of form / travel / rest), both teams FBS = 1u, its own record, never pulled. Judged at 50, cut under -5%. The 10-15 gap lost everywhere; 15+ is the only band that paid.
- Lock 1u. No college favorite LEAN at -150..-130. College dogs sit out the daily floor (7-12 / 6-11 in the replay).
- Early: off a bye 1u, hammered ½u, blew somebody out ½u (wins over FBS teams only). 'Engine likes it, line moved away' is PAUSED (live 3-6). A college read counts 5 pts max in the early spots.

7/12 🏀 NBA

- The read: the rating + a roster read (tonight's men at last season's value) blended over the first 10 games. NBA is owner-on - never shut out.
- Favorites read off the books' price (the own read loses to the close every season), moved by the study list (cap 4): a top-5 man back after 2+ missed -3 | played OT last game -3 | own top-2 man out -2 | opponent's top-2 out -2 (books over-move) | 2+ more days of rest -2 | 4+ possessions faster -2 | first home game after a 4+ trip -2 | across a star off a dud -1 | off a 20+ win +2 | at a home dog in its games 31-60 +2. (A favorite that wins inside also gets +1 on its win %.)
- No NBA Lock in a team's games 1-5 (-12.1% blind).
- 🏀 NBA FAVORITE PLAY 1u: -150..-101 whose study weights net positive (62-25, +23.1% on 87).
- Dog gate 4, no own-read rule; the score is its named reason. 1u if its own read beats the price, else ½u.
- Dog angles (2 each): lost the last meeting by 15+ +2 | moneyline longer than its spread says +2 | the favorite played OT last game +2 | 12+ game slate +2 | home dog in games 6-10 +4, 11-20 +2 (+100..+200) | rested vs a back-to-back +3
- Dog fades (-2 each): home dog in games 31-60 | off a 15+ win | total 8+ under the league | favorite on its 4th+ straight road game | our first home game after

a 4+ trip | favorite off a 20+ win | blew a 10+ halftime lead last game | played OT last game | scored 130+ last game | favorite scored 95 or less | won 3+ straight | rated near even but +200 | favorite 2+ more days rest | opening 15 games, a .550+ team last year | the favorite's next game is soft | lost a 20+ minute man (games 4-10) | market faded it 8+ pts over its last 3 | cold (under .300 after 8+, or a 5+ skid). Won 2+ close games -3; we won and they lost -2.

- 🏀 NBA FLOOR ½u: the best NBA dog by score (0+) when the board has no NBA unit play (84-110, +16.5% on 194).
- Spreads (leans): a favorite laying 3.5-12.5 in its first 20 games -3 on its rank.
- Early: the NBA opening-line play ½u (part 10).

8/12 🏀 COLLEGE HOOPS

- Efficiency ratings (tempo-free, opponent and venue adjusted, from our own box scores) carry this sport - the raw rating is weak in hoops. A non-D1 game moves nothing (the 11/4 South Dakota fix).
- No Dog of the Day, no runner-up dog play, no daily floor in college hoops.
- THE DOG READ = the study weights (capped ±6) + the ratings (efficiency backs the dog +2, +2 more at a 15-pt gap, 5+ pts against it -2) + facts (defense-first vs a bottom-third offense +2 | whole team vs 2+ out +1 | opponent's star in his 2nd game back +1 | off an upset of a top-25 +1 | Jan-Mar offense-first / transfer-heavy / first-year coach / cold shooting -1 each | 3-point-hot on the road -1).
- 🐶 DOG SLOT: read 4+ with both teams 4+ games - up to 2 a night, 1u, 2u at 6+ (read 4+ went +16.5% on 600, every season 2018-25 up).
- 🔍 GAP TWIN: efficiency read 15+ over the no-vig, backing the dog, no study fade, both teams 4+ games = 1u, 2u at a dog read 6+ (102-103, +22.6% on 205).
- 💲 FAVORITE SLOT ½u: -120..-150, weighed read 56%+ and 1-5 pts over the price, own read over it too, no favorite fade, both teams 3+ games. Each slot judged at 50 - under -5% = off.
- Dog fades: star's 2nd game back -3 (those dogs -51.1% on 87) | Feb-Mar also-ran (5+ back) vs a contender -3 (-31.2% on 664) | rubber match -2 | 5+

game skid -2 | top scorer missed the last game -2 (-1 more at 5+ straight) | a regular out 4+ straight -2 | road-cold (6+ short of its number on the road) -2 | top-25 road dog Nov-Dec -2 | road dog back from a neutral-site event -2 | lucky from three -1 | won 2+ close games -1 | out-rebounded by 6+ -1 | we lost and they won -3

- Dog boosts: home dog vs a top-25 road team Nov-Dec +2 | out-rebounds by 6+ +1, 8+ +2 | the favorite lost its last +2
- Favorite fades: away from home Nov-Dec -3 | efficiency read 3+ under the price -3 | star's 2nd game back -3 | first-year coach Jan-Mar -3 | own top scorer missed the last game -2 | turnover-prone -2 | lucky from three -2
- The early road dog fade (-2, games 1-3) is OFF since 11/9 - its own judge tripped (those dogs +23.5% on 53).
- Locks: none until both teams have 3 games, ½u through game 10. Spreads are leans only (a 12+ home favorite -3 on its rank, -8% ATS).
- Injuries: the 2+ out / 4+ questionable no-units rule. Early: ⛰️ the over at altitude ½u (part 10).

9/12 🏒 NHL (+ MLB & TENNIS)

- The read: the books' line + back-to-back and rest only. Moneylines only - no puck lines.
- Dog gate 6 (6+ won 48.7%, +12.8% on 542, 7 of 7 seasons); the score is its named reason. Its own read can't sit under the no-vig number. Fallback Dog: the best score over 3 with its own read 2+ pts over the no-vig (in practice the rested dog vs a favorite on a back-to-back).
- Dog angles: rested vs a back-to-back +3 | both teams lost their last +3.5 | out-shooting them +3 | puck luck against them (PDO 985 or under) +2 | goals say they're better than the record (games 5-20) +2 | out-hitting people +2 | the favorite off an OT loss +2 | starts its #1 goalie and the favorite doesn't +2 | money came in on it +2 | off an OT win +1 | facing a team that allowed 40+ shots +1 | sharp money against the tickets +1 | the other goalie's hot run +1
- Dog fades: money ran away -4 | played last night, the favorite didn't -3 | lucky (PDO 1015+) -3 | out-shot -3 | its goalie's hot run -2 to -3 | +200 or longer -2,

+220 or longer -3 more | 3+ game win streak -2 | off an OT loss or a shootout win -2 | allowed 40+ shots last game -2 | facing a slumping goalie -2 | rested dog whose #1 sat 2+ straight -1

- Favorite: no-vig + ½ pt per dog-score point the other way (cap 4) | first 2 weeks of the season -3 | rested vs a team on its 3rd game in 4 nights +1.5 | its own goalie slumping +1.5 (the books over-move) | allowed 40+ shots last game -1 | wins 1st periods past its record -1. Its weighed read has to beat the price; no hockey favorite LEAN at -150..-130 (191-185, -12.5%).
- 🥅 Goalies: the confirmed / likely starter (Daily Faceoff); nobody confirmed = the projected one (#1 on a rested night, the other goalie on a back-to-back - right 63% vs 43% for 'last starter'), and the card says 'projected'. Goalie form = save % over his last 5 starts.
- Dog size: 2u at +100..+150, ½u at +151..+350. Cold hockey dogs get NO fade (they won).
- ⚾ MLB (when in season): drought favorite (no runs in 12+ innings) +3 | doubleheader game-2 dog +1 | +200..+249 dog -4 | out-scoring the favorite lately +1.5 | late rally +1.5 | unluckier +1. Live baseball paused.
- 🎾 Tennis: posts 8 AM, no units. Live tennis plus money only on SUPER value - a 65%+ pre-match favorite now at plus money with a 10%+ edge.

10/12 ⏰ EARLY VALUE PLAYS

- Posted days ahead at the first FAIR number (after both teams' last games) - within 3 days of it, never on game day. Checked every hour, no weekly cap, no phone ping.
- The spots are WEIGHTS on the engine's read: bye +4, East-West +3, blowout +2, hammered +2, Monday night +1.5; fades (ice cold, coach's first season, losing streak) -4 each. It posts only if the total is 5+ pts and the own read is no more than 1 pt under the price.
- Dogs +100..+350 only. Never a side with a key player out or questionable.
- Each spot keeps its own record: 🛌 off a bye (NFL, college) 1u | 🔨 hammered early (the money ran 4+ pts to the dog since the first fair price) ½u | 🏈 Monday night NFL dog ½u | ✈️ East Coast NFL team flying West ½u | 💥 blew

somebody out (college only, FBS wins only) ½u | 🧠 engine-vs-the-line (college) PAUSED since 10/24.

- 🌍 Europe morning NFL under ½u: every NFL game in Europe before noon ET, posted from 6 PM PT the night before (17-9 since 2018). Under 50% after 15 = off.
- ⛰️ College hoops over at altitude ½u: a home floor 4,000 ft+, the points read 2+ over the total, posted the night before (445-318, +11.1% 2018-26). Judged at 50.
- 🏀 NBA opening line ½u: the whole-model read (37 factors) at the first price, best side a night, either side -150..+300, 4+ pts over the open (334 bets +13.4% blind 2023-25). Back to ½u 11/3 - the 2020-23 seasons didn't back 1u. Judged at 40.
- A pulled early play can post again 12+ hours later if it clears again. On game day the board can take the same side too (both count) - never the other side.

11/12 💰 SIZING & THE BOARD

- Board at 8 AM PT on game day (once the news is in; at the latest 3 hours before the first game). It fills to 5 picks with leans. 1u = $1,000 flat on the $100,000 roll (+$500 per full $50,000). Max play 5u.
- 🔒 LOCK: a 56%+ read whose own read (weighed read where there is one) beats the real price, -150 to +125, never -119..-101 (46% in the replay); the likeliest winner first. 2u - but 1u when its own read is 7+ pts over the market (the biggest gaps were the worst), 1u in football, ½u in college hoops games 3-10. Thin edge (under 3%) = not a Lock (a ½u value play). Backup Lock ½u. Nothing clears = 'No Lock of the Day' - never forced. The underdog Lock is OFF.
- 🐶 DOG: hockey & football 2u at +100..+150, ½u at +151..+350 (the short Dog 85-68, +22.1%; the long one 15-33). NBA 1u (own read over the price) or ½u. None clears = no Dog. A dog found later with 3+ pts more value takes the spot; the old Dog becomes a ½u value play.
- 🔥 VALUE PLAYS ½u: favorites only, moneyline only, a 56%+ read with its own read beating the price. Runner-up dog ½u (passed every Dog check, one

a day, not college hoops). NBA favorite play 1u. 🔍 gap 1u. College hoops slots ½u / 1u / 2u.

- DAILY FLOOR: a board short of 2 unit plays fills with ½u floor dogs (best dog score - NFL, NHL, MLB, thin edges), so every day has at least one unit play. NBA floor ½u.
- ⭐ ENGINE'S BEST PICK: one label a day on the best unit play - Lock > Dog > NBA favorite > 🔍 / runner-up > college hoops slot / NBA floor > night pick > floor > value play > backup Lock, then size, then edge - raised to 1u at least. Since 11/5 only on a play whose read beats its price; none = no ⭐. A day with no unit play: the surest side -150..+350 at 1u, same rule.
- 🟡 LEANS: no units, who-wins calls (STRONG at 53%+, SLIGHT under), 2 a sport max, the big games first.
- Hard rules: never past -150, no dog past +350, no puck / run lines, one pick a game, no posted parlays (🧩 build your own). Graded cards stay up all day, into the results at 1 AM PT.

12/12 🔎 THE CHECKERS & THE RECORDS

- Slate check: every game named, priced and read by the engine - a problem holds the 8 AM post (re-pulls 8:12, 8:32) until 8:30 PT.
- Factor check: every study's data and every injury feed loaded - a missing piece holds the board.
- Pick check (every pick, since 11/4): game not started | a read and a price | price under 6 hours old | injury data on BOTH teams | our starting QB not quietly out / questionable | no unit play waiting on news | today's goalie page | college: we hold every game of both teams | never a non-D1 opponent | every named reason TRUE ('the stronger team' = rated higher without the home edge and read 50%+). A fail pulls the pick.
- Rule check: -150 cap, no puck / run lines, one pick a game, Lock / Dog carry units, the Dog's price cap, units only when the read beats the real price. A sizing check reports every size.
- Injury hold & watch: a key questionable starter holds a unit play; news after posting shows on the card - a posted pick never changes without the owner's

OK.

- Live plus money: no units, never against a pick we have on that game, MLB paused, a DraftKings-only price needs a second book.
- TWO RECORDS, NO OVERALL: 💰 UNIT PLAYS (every Lock, Dog, value play and early play - W-L, units, ROI; BY SPORT is unit plays only) and 🟡 LEANS. Live plus money and tennis keep their own; each early spot, the 🔍 play and the ⭐ have their own lines.
- How a study gets in - five checks: (1) fair - only what's known before the game; (2) blind - it holds on seasons it never saw; (3) most seasons the same way; (4) this season agrees; (5) a second check (the open, the spread, the dose). Then it's a WEIGHT sized by how sure, inside the ±6 / ±4 caps - never a trigger - with a tracker tag judged at 50. A weight that stops working comes off (the college hoops early road dog, 11/9).

From the engine's code on main, 11/11. The learned numbers retune every day.